Field Crops

India has one of the largest agricultural sectors worldwide, which is the primary source of livelihood for ~55% of India's population. India has the world's largest cultivated land under wheat, rice, and cotton. India is the largest producer of milk, pulses, and spices in the world. It is the second-largest producer of fruits, vegetables, tea, farmed fish, cotton, sugarcane, wheat, rice, cotton, and sugar. The agriculture sector in India holds the record for second-largest agricultural land in the world, generating employment for about half of the country's population. Thus, the farming community comprise an integral part of this sector to provide food security with a means of sustenance.

During the last 9 years (2014-2023), a total of 2279 varieties of field crops have been released in India under the aegis of ICAR, which include 1092 of cereals, 343 of oilseeds, 343 of pulses, 137 of forages crops, 279 of fibrecrops, 68 of sugarcane and 17 of potential (underutilized) crops. These varieties include 1888 climate resilient varieties (83%) tolerant to one or more biotic and/or abiotic stresses. Out of these 1888 varieties tolerant to one or more biotic and/or abiotic stresses, 407 varieties have been developed specially for extreme climate including flood/ water submergence/ water logging tolerance (73), drought/moisture stress/ water stress tolerance (220), salinity/ alkalinity/ sodic soils tolerance (52), heat stress/ high temperature tolerance (49), cold/ frost/ winter chilling tolerance (13) using the precision phenotyping tools. In horticultural crops also, 22 climate resilient varieties have been developed. This includes varieties of potato (5), tomato (2), spinach (1), radish (1), cassava (4), sweet potato (2), greater yam (1), white yam (1), coconut (3) and taro (2).

Further, to tackle the widespread problem of malnutrition in globally, concerted efforts have been made to develop varieties rich in various essentially required micro-nutrients (Protein, Fe, Zn, Vit. A, Vit. C, oleic acid, anthocycnin, lysine, tryptophan, removal of anti-nutritional factors like erucic acid, glucosinolates, KTI) for proper growth and development of the body and as a result 124 field crops and horticultural crops varieties have been released and notified during past nine years [Rice (10), Wheat (41), Maize (20), Pearl millet (9), finger millet (4), little millet (1), Lentil (2), Mustard (8), Linseed (1), Soybean (7), groundnut (3) and 18 Horticultural crops]. Concerted efforts for seed production have been made to take the biofortified varieties in general cultivation, which have resulted in coverage of more than 10 million hectares during 2022-23. Biofortified varieties have more than 50% share in total wheat breeder seed indents.

1. Implementation of GAP

The ministry of Agriculture and Farmer's Welfare and Processed Food Products Export Development Authority (APEDA), Govt. of India implements the GAP. APEDA adopts the approach of developing product specific clusters in different agro- climatic zones of the country to help in dealing with various supply side issues through Good Agriculture Practices addressing issues such as, soil nutrients management, higher productivity, adoption of market-oriented varieties etc.

The Quality Council of India (QCI) is conducting implementation of India Good Agricultural Practices (INDGAP) standard and certification of projects (https://www.qcin.org/india-good-agriculture-practices).

2. Major constraints in Implementation of GAP

There are certain challenges in implementation of GAP both at farmers level and market level. At farmers level, almost there are nearly 80% marginal and small farmers. Resource at farmer's level is very low – agricultural implements, protective gears etc. Farmers' use own seed/planting material, especially, less use of High Yielding Varieties. Non availability of genuine inputs such as, bio fertilizers, fertilizers, over dependence on local traders for information leading to overuse of agrichemicals. At market level, small farmers selling produce at local level to avoid loss, less transportation Infrastructure at rural areas causing inaccessible to markets and less information on market intelligence are some of the factors. The initiation of eNAM (National Agricultural Marketing) Platform will facilitate marketing of agricultural produce having GAP Certification.

3. Quality Council of India is associated for Certification of GAP

Food Safety Management Systems, Product Certification, Inspection, INDGAP certification, India-GHP Certification, India-HACCP Certification etc. It has following broad objectives:

- To develop training material on INDGAP in consultation with Agricultural and Processed Food Products Export Development Authority (APEDA).
- Design and operationalize Capacity Building program in the form of workshops to disseminate the INDGAP amongst various stakeholders across various commodities.
- To implement INDGAP Standard and certification of required projects as per APEDA mandated crops.
- To implement the benchmarking process as per global GAP requirements.
- To coordinate with concerned ICAR institutions for the development of SOP for GAP on different crops in a time bound manner

4. Potential of GAP

There are more than 5000 Farmer Producing Oraganizations (including FPOs)/ Farmers Producer Companies (FPCs) by various schemes and NABARD and through Cooperatives/Societies etc. In India, majority of these FPOs with 100 to 1000 farmers. Under Small Farmers Agribusiness consortium (SFAC) there are nearly 910 FPOs with 8.92 lakh farmers in 29 states. The target is to create 10,000 FPO/FPCs in five years and further promoting these FPOs for GAP has a great potential at global level.

5. Role of Organizations in GAP implementation

The technology development for crop production and management of Horticulture crops is mainly undertaken by the public funded research institutions in the country for several decades. The Indian Council of Agricultural Research (ICAR) is the apex body for co- ordinating, guiding and managing research and education in agriculture including horticulture, fisheries and animal sciences in the entire country. With 113 ICAR

institutes (Crop or commodity based) and 71 agricultural universities spread across the varying agro climatic regions of the country, NAREES (National Agriculture Research Education and Extension) is one of the largest national agricultural systems in the world. Many INM technologies including micro irrigation/fertigation methods and bio-control strategies to reduce use of chemicals in crop production were developed and popularized. ICAR has developed the production protocols of GAPs for several crops, vegetables, fruits and spices and medicinal crops which are ready to be adopted.

The Agricultural & Processed Food Export Development Authority (APEDA) has focused on collaborative approach for capacity building of various stakeholders involved in the development of export enhancement of fruits by creating Agri export clusters for different commodities in India. Thirteen Clusters are identified for district(s) like potato in Punjab and Uttar Pradesh; isabgol in Rajasthan; orange, pomegranate and grapes in Maharashtra; banana in Tamil Nadu; mango in Uttar Pradesh; rose and onion in Karnataka and fresh vegetables in Uttar Pradesh. The Quality Council of India (QCI) in partnership with the India's leading industry bodies,- the Confederation of Indian Industries (CII), the Federation of Indian Chambers of Commerce and Industry (FICCI) etc has launched accreditation and certification schemes for the identified sectors such as Food Safety Management Systems, Product Certification, Inspection, INDGAP certification, India-GHP Certification, India-HACCP Certification etc.

6. Good Agricultural Practices for Rice production

A. Water

- Water for growing rice shall be from source that environment is safe from hazardous substance contamination.
- The construction of separate channels to move water to and from each field greatly improves the control of water by individual farmers.
- Field channels allow water to be delivered to the individual seed beds separately and the main field does not need to be irrigated until it's time to plant in the main field.
- In addition, the ability to control water to your field is important when you need to retain water (especially after applying fertilizer so nutrients are not lost) or when you need to drain the field for harvest.
- Construction of individual field channels is the recommended practice in any type of irrigation system.
- Continuous flooding of water generally provides the best growth environment for rice. After transplanting, water levels should be around 3 cm initially, and gradually
  increase to 5–10 cm (with increasing plant height) and remain there until the field is drained 7–10 days before harvest.
- For direct wet seeded rice, keep the soil moist but not saturated from

sowing till emergence, to avoid seeds from rotting in the soil.

- field should be flooded only once the plants are large enough to withstand shallow flooding (3-4 leaf stage).

B. Planting Area

- Timely planting into a well-prepared seedbed will help produce a fast growing, uniform crop that will have higher yields and better competition against weeds and other pests. The best time to plant depends on locality, variety, weather, water availability, and the best harvest time. Planting at the same time (or within a two- week window) as the neighbouring fields can help to minimize insect, disease, bird, and rat pressure on individual fields.
- A well-prepared land controls weeds, recycles plant nutrients, and provides a soft soil mass for transplanting and a suitable soil surface for direct seeding.

C. Seeds and Sowings

- Use healthy seeds of a locally adapted variety to get a good crop that has high yield potential and a good market price.
- Seed must be grown, harvested, and processed correctly for best yield and quality results. The number of plants established and its seedling vigor will affect the competitiveness of the crop against weeds, and determine the final yield potential.
- Direct seeding requires 60–80 kg of seeds per ha, while transplanting only requires 40 kg per ha, at 2 plants per hill.

D. Planting

- Rice crops can be either direct seeded or transplanted. It can be manual with labour or with transplanters.
- The seedlings should be planted at an ideal depth of 4-5 cm at spacing of 15cm x 20cm or at random at 15 cm spacings.
- It is ideal to leave alleyways every 2 m for interculture operations, movement of labour, controls pests and the better aeration results in more productive tillers.

E. Intercultural Operations

- Ploughing and harrowing in fallow should be undertaken at least 10–14 days apart or after rain.
- First weeding begins within 2–3 weeks after establishment and the second in another 2–3 weeks. Weeding before fertilizer application is ideal.
- Identify the weed correctly and use the appropriate herbicide as recommended on the label and Spray when the weeds are small.

F. Fertilizer and Pesticides

- Use organic fertilizer (manure, compost, straw, husk, plant leaves) whenever possible, especially in nurseries.
- Apply fertilizer according to soil type and expected yield. As a guide, a 2 t/ha yield on clay loam soil will require 20 kg N and 5 kg P. Sandy soils may require another
  10–15 kg K. Double these recommendations for a 3 t/ha expected yield.
- Apply all P, K, and 10% N evenly and incorporate just before seeding or transplanting. For direct seeded broadcast crops, it is okay to apply 10–14 days after establishment when there is water in the field.
- Apply remaining N (urea) in 2 equal portions at 30 days and 50–60 days (panicle initiation) after emergence.
- In established crops, apply chemical fertilizer only in standing water and evenly across the whole field.
- Do not apply high rates of fertilizer for traditional varieties as they may have limited response and cause lodging.
- If application of pesticide is required, it has to be as per the recommendations of ICAR, SAU's and Agriculture Department.
- Do not use the hazardous substances which are prohibited for agricultural use.
- In case of rice produced for export, do not use pesticides which are prohibited by the importing country.

G. Pre-Harvest Quality Management

- Crop Uniformity and purity has to be maintained at all intervals. The, off types need to be removed immediately. The quantity of off type rice plants shall not exceed 3%.
- Crop has to be regularly observed for diseases and pests and suitable control measures to be adopted.
- Produce shall not have any plant disease and damages by insect greater than 10%.

H. Harvesting and Post Harvest handling

- If threshing and/or harvesting is applied, equipments shall be properly cleaned and threshing process shall be handled carefully in such a way that it does not introduce any admixing grain to the produce. If the machine has been previously used to harvest or thresh other rice variety, it shall be cleaned to get rid of all remaining grains.
- If produce is not sold as wet paddy, it must be dried to reduce moisture within 24 hours after harvested.
- Drying practice shall not introduce any breakage to grain in such a way that the result of milling quality test of paddy is lower than the criteria required in Thai Agricultural Standard for each type of rice.
- The moisture of paddy shall not exceed 15% for traded rice and 14% for stored rice.

I. Transport, Holding and Storage

- Equipments, containers and carriages for transportation and storage shall be cleaned and able to prevent quality deterioration of produce and prevent from contamination of substances that may be harmful to consumers, and also from inducing of admixing grain.

- Grain collecting and storage room shall be managed in such a way that the condition is hygienical clean and well ventilated. They are able to prevent contaminations and inducing of admixing grain.
- If more than one variety of rice is handled or stored, management in storage shall be in such a way that admixture of grain between different varieties is prevented.
- The dried seeds to store in jute bags, properly stitched, labelled and stacked uniformly in manner with walking space on all four sides.
- The stacks to be away from walls and moisture and properly covered.
- Regular fumigation and rodent control methods to be adopted to prevent damage and losses.

J. Personal Hygiene

- The pesticides and herbicides are poisonous; if they are not used properly, they can cause health and environment problems. Label them clearly and keep them out of children's reach.
- Always use protective clothing when spraying. Do not wear raincoats when spraying as this increases sweating. Personnel and labour shall wear clean clothing suitable for the operations with which they are involved.
- Personnel working in the production area should refrain from non-hygienic behaviour, such as smoking, spitting, and chewing during the operations.
- Visitors who are allowed to enter the production area shall receive prior permission and adhere strictly to the personal hygiene instruction

K. Record Keeping and Traceability

A record in proper format to be maintained for seed sources and time of sowing/ planting, water source, soil preparation, the off-type rice removal, the survey of plant damages by pest and pest management, Date and time of application of pesticides, Time of harvesting and threshing, drying for moisture reduction and paddy packaging and storage.

Good agricultural practices (GAP) of important cultivated forage crops

Crops and Varieties	Sowing/planting time	Seed rate & spacing	Fertilizer management	Water management	Weed management	Harvesting
Sorghum PC-6, PC-9, PC-23, MP Chari, UP Chari-1, UP Chari-2, HJ-541, CSV 32F, CSV 33MF,	Single cut: June-July, Multi-cut: March-April	25-30 cm row to row seed rate :35-40 kg/ha (Bold seeded), 25-30 kg/ha (small seed)	FYM: 10 t/ha, Single cut varieties: 60:30:30 kg N: P_2O_5 :K_2O/ha as basal and 30 kg N/ha as top dressing.	Rainy season crop: 1-2 irrigations Summer sown crop: 5-6 irrigations	Pre emergence application of atrazine @ 0.5 kg a.i./ha in 450 litres of water is effective. One	Single cut: 60-75 days after sowing (DAS) *i.e* 50% flowering, Multicut: first cut

			Multicut: 70:30:30 kg N: P_2O_5: K_2O/ha as basal and 50 kg N/ha as top dressing		hoeing through weed-cum-mulcher at 3-4 week crop stage.	at 40-45 DAS and subsequent cut at 30 days interval
Pearl millet Single cut: Raj Bajra Chari-2, CO-8, APFB-2 Dual: AVKB-19 Multi cut: Giant Bajra, Proagro-1	Summer: March to mid April. Monsoon season: first fortnight of July	25 cm row to row using a seed rate of 10-12 kg/ha	FYM: 10 t/ha 50:30:30 kg N: P_2O_5: K_2O/ha as basal and 30 kg N/ha as top dressing. In rainfed 20-30 kg N/ha coinciding with rain at 30-35 days stage	Rainy season crop: 1-2 irrigations Summer sown crop: 4-5 irrigations	Pre emergence application of atrazine @ 0.5 kg a.i./ha in 450 litres of water is effective. One hoeing through weed-cum-mulcher at 3-4 week crop stage.	Single cut: 55-60 DAS (initiation of flowering) Multicut: first cut at 40-45 DAS and subsequent cut at 30 days interval
Maize African Tall, Vijay, Moti, Jawahar Composite, VL-54, APFM-8, J-1006	Summer: February to March. Monsoon season: Beginning of rains June-July	30-40 cm row to row, using a seed rate of 40-50 kg/ha	FYM: 12-15 t/ha, 80-100 kg N + 40 kg P_2O_5/ha. 15-20 kg $ZnSO_4$ ai/ha	Rainy season crop: 1-2 irrigations Summer sown crop: 5-6 irrigations at 10-12 days interval	Application of atrazine @ 0.75-1.0 kg a.i./ha in 450 litres of water is effective. One hoeing through weed-cum-mulcher at 3-4 week	Silk stage (60-75 DAS) for fodder purpose which continue up to milk stage

					crop stage. Pre-emergence	
Cowpea EC-4216, UPC-5286, BL-1, BL-2, UPC-618, UPC-622,	Sowing time extends from March to	Sowing should be done in lines at inter row spacing of 25-30 cm	20 kg N and 60 kg P_2O_5/ha at sowing. 20-40 kg sulphur/ha	Monsoon season: Crop doesn't require irrigation	PPI of Fluchloralin @ 0.75 kg a.i/ha. One manual	Rainy season crop: at 50-60 DAS at 50%

GFC-1, GFC-2, GFC-3, IL-1177, MFC 8-14	middle of July.	with seed rate of 35-40 kg/ha	in sulphur deficient soils	except in case of long dry spell. Summer: 6-7 irrigations at 8-10 days interval	weeding or hoeing with weeder-cum-mulcher at 3 week crop stage.	flowering. Summer season: 70-75 DAS
Cluster bean Bundel Guar-1, BG-2, BG-3, Maru Guar, Guara-80, HFG-110, HFG-156	Summer crop: March-April, Monsoon seaon: June-July. Winter: October-November	25 cm row to row spacing using a seed rate of 30-35 kg/ha. Under dry land 25-30 kg/ha at 30 cm spacing	20 kg N and 50 kg P_2O_5/ha at sowing	Monsoon season: Crop doesn't require irrigation except in case of long dry spell. Summer: 3-4 irrigations	Pre-plant incorporation of Nitralin @ 0.75 kg a.i/ha. At 3-4 week crop stage hoeing with weeder cum mulcher.	At bloom to pod formation stage (60-75 days after sowing)
Oat JHO-822, JHO-851, JHO-2012-2, Bundel Jai 2000-4 HFO-114, Kent, OS-6, OS-7, JHO 2015-4, HFO-607, HFO-707, JO-05-304	Mid October-end of November in North West to East zone.	Sowing preferably done by pora/kera behind the plough. 80-100 kg/ha. 30 cm row spacing for profuse tillering varieties	15 t FYM/ha before 15-20 days of sowing. In case of single and two cut 80 kg N and 40 kg P_2O_5/ha, in case of multicut 100-120 kg N, 40 kg P_2O_5 and 40 kg K_2O/ha.	4-5 irrigations including the pre-sowing irrigation. For multi-cut varieties 7-8 irrigations are required	Weeding with weeder cum mulcher at 4 week crop stage followed by application of 2,4-D @ 0.50 kg a.i at 30-35 DAS.	Single cut: 50% flowering stage Double cut: First cut at 60 days followed by second cut at 50% flowering stage. Multicut: First cut at 60 days,

						second cut at 105 days and third cut at 50% flowering

Berseem Mescavi, Wardan, BL-1, BL-10, BL-2, Bl-22, JB-1, JB-2, JB-3, BB-2, BB-3, BL-43, BB-5, BB-6, BB-7	Mid October (Punjab, Haryana and U.P) November (Gujarat and W.B)	Optimum time sowing: 25 kg/ha. Early sowing: 15-20% extra seed rate. Lowland Rice: 35 kg/ha	20 kg N/ha, 80-90 kg P_2O_5/ha and 30-40 kg K_2O/ha at sowing,	16-18 irrigations in 10-12 days interval	Field infestation of chicory minimized by treating with 10% common salt and deep summer ploughing. Imazethapyr @ 0.10 kg a.i/ha as PPI was effective	First cut at 55 DAS and subsequent cuts at 25-30 days after previous cut.
Lucerne Sirsa-8, Anand-2, Anand-3, RL-88, Co-1, T-9	End of September to early December but best time is Middle of October.	Broadcasting: 20-25 kg/ha, line sowing 12-15 kg/ha. It can also sown with seed drill at 25-30 spacing	20 t FYM/ha every year for perennial crop. Seed inoculation with *Rhizobhium meliloti*. 20 kg N, 60-75 kg P_2O_5 and 40 kg K_2O/ha	Crop requires 15-20 irrigations in a year along with pre sowing irrigation. Early stage frequent irrigation required	Fist weeding at 20-25 DAS. pendimethalin @ 1.2 kg a.i/ha A Pre-emergence or spot application of diquat @6-10 kg/ha as Post emergence (5-10 DAS) controls *Cascuta*	First cut at 50-55 DAS and subsequent cuts at an interval of 25-30 days. In a year 8-10 cuts.
Napier Bajra Hybrid IGFRI-3, IGFRI-6, IGFRI-7 and IGFRI-	Planting through rooted slips at any time of the year except in	Sole crop: 35000 rooted slips or stem cuttings/ha at 75×50 cm spacing.	20-25 t FYM/ha. At sowing basal dose of 60 kg N, 50 kg P_2O_5 and 40 kg	During monsoon no need of irrigation except long dry spell.	Regular hand weeding/hoeing ensures good aeration	First cut at 60-65 DAS and subsequent cuts at an interval

10, NB-21, CO-1, CO-2, CO-3, CO-4, CO-5, CO-6 PBN-83,	winter. Irrigated: February Rainfed: July-August.	Inter crop: 20000 rooted slips/ha at 100×50 cm spacing	K_2O/ha applied prior to sowing. 30 kg N/ha after each cut	During march-may crop require regular irrigation at 15-18	and crop growth.	of 25-30 days. In a year 6-8 cuts can be taken.

DHN-6, APBN-1,				days interval. 10-12 days interval in summer		
Guinea Grass Hamil, BG-1, BG-2, CO-1, CO-2, PGG-1, PGG-9, PGG-14, PGG-19 and PGG-101.	Irrigated: Mid February-July. Rainfed: Monsoon season.	Seed rate of 3-4 kg/ha for sole crop. 40000 rooted slips (sole) and 20000 rooted slips (intercropping). 20-25 days old seedling or rooted slips at 50×50 cm for sole and 150×50 cm for intercropping.	20-25 t FYM/ha. At sowing Basal dose of 60 kg N, 50 kg P_2O_5 and 40 kg K_2O/ha applied prior to sowing. 40 kg N/ha after each cut	During monsoon no need of irrigation except long dry spell. During march-may crop require regular irrigation at 15-18 days interval. 10-12 days interval in summer months.	Regular hand weeding/hoeing ensures good aeration and crop growth.	First cut at 60-65 DAS and subsequent cuts at an interval of 25-30 days.

7. Safe and Judicious Use of Pesticides as per GAP

In India, under the All-India Network Project on Pesticide Residues (AINP-PR), 18 laboratories perform field studies in different agroclimatic zones as per GAP for generating persistence and residue data for pesticide evaluation which help in determining the safe waiting period, pre-harvest intervals and recommending safe and judicious use of pesticides among farmers. The data obtained through the Project aids the regulatory authorities of the country in registration of pesticides and approval of label claims on different pesticide-crop combinations for commercial use in the country and the fixation of their Maximum

Residue Limits (MRLs).

8. Surveillance of Pesticides Residues

The ICAR implements the surveillance programme of pesticide residues in India through the Monitoring of Pesticide Residues at National Level (MPRNL). Presently, 35 laboratories representing Ministry of Agriculture, ICAR, Ministry of Health and Family Welfare, Ministry of Environment and Forest; Council of Scientific and Industrial Research (CSIR), Ministry of Chemical and Fertilizer, Ministry of Commerce and State Agricultural Universities are involved in the project. The participating laboratories are entrusted with the regular collection of food commodities namely vegetables, fruits, spices, herbs, cereals, meat, egg, tea, milk, fish and marine products collected from the retail outlets, Agricultural Produce

Marketing Committee (APMC) markets, farm gate, organic outlets and surface water from different parts of the country and monitoring the presence of pesticide residues. The information related to the pesticides detected, their levels, samples above MRLs-off- label pesticide detection, banned pesticides detection, etc. is shared with the state governments for promoting the safe and judicious use of pesticides among farmers and for Integrated Pest Management (IPM) interventions. The data is also used for assessing consumer risk. The monitoring data on spices generated under this scheme has also been submitted to FAO/WHO JMPR for fixing 20 Codex MRLS (CXLs) on spices.

9. Accreditation of Laboratories

It is ensured that the laboratories involved in pesticide residue testing are accredited by National Accreditation Board for Testing and Calibration of Laboratories (NABL)as per ISO/ IEC 17025:2017 to generate quality data.

10. Capacity Building

Regular training programmes have been organized for providing systematic guidance on determination of pesticide residues in food commodities and updating the knowledge and skills of the researchers and apprising them about international standards and guidelines (such as SANTE and Codex) in the field of pesticide residue analysis. These trainings help the technical personnel in the accurate determination of pesticide residues in various food commodities collected from different sources throughout the country. The scientists/researchers attending

these trainings gather skills and become proficient in the analysis of pesticide residues and further disseminate the knowledge to the peers working in the field of pesticide residues. The trainings have helped build up a robust system and skilled workforce for pesticide residues testing across the country.